CATHOLIC JUNIOR COLLEGE
JC2 PRELIMINARY EXAMINATIONS
Higher 2

Answers

CANDIDATE NAME			
CLASS	2T	**INDEX NUMBER**	

BIOLOGY

9648/03
25th August 2016
2 hours

Additional Materials: Writing Paper

READ THESE INSTRUCTIONS FIRST

Write your index number and name on all the work you hand in.
Write in dark blue or black pen on both sides of the paper. **[PILOT FRIXION ERASABLE PENS ARE NOT ALLOWED]**
You may use a soft pencil for any diagrams, graphs or rough working.
Do not use staples, paper clips, highlighters, glue or correction fluid.

There are two sections in this paper.

Section A]
Answer all questions

Section B]
Answer all questions. Answer each part on a **separate** piece of paper.

At the end of the examination, fasten all work securely together.
The number of marks is given in brackets [ ] at the end of each question or part of the question.

For Examiner's Use	
Section A	52
1 [13]	
2 [13]	
3 [14]	
4 [12]	
Section B	20
5a [6]	
5b [6]	
5c [8]	
TOTAL	72

This document consists of 22 printed pages and 0 blank page.

[Turn over

Section A

Answer **all** questions in this section.

1 Human Growth Hormone is important to augment normal growth and development in the treatment of individuals with dwarfism. Fig. 1.1 shows how human growth hormone can be produced via expression of recombinant DNA in *Escherichia coli* host cells.

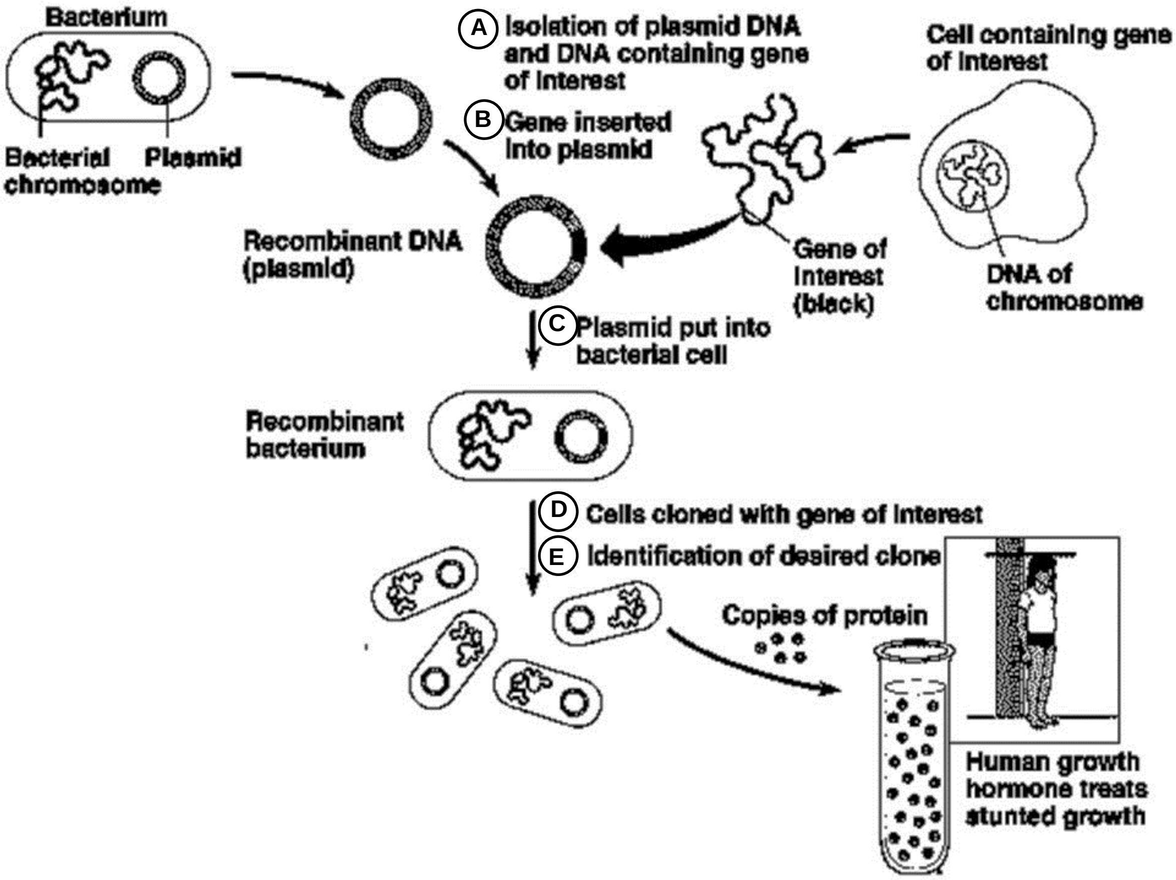


Fig. 1.1

(a) Explain what is meant by recombinant DNA.

………

………[1]

1. **Genes from two different sources / organisms** are **combined** *in vitro* into a single plasmid **/ OWTTE** (accept contextual answers).

[L1]

(b) Name the process required in the following procedure in Fig 1.1.

C : ………………………………**Transformation**

E : ………………………………**Selection / Screening**

[1]

(Both correct – 1 mark)

[L1]

(c) The gene of interest cannot be taken directly from DNA of chromosome but require additional processing in order to produce functional protein.

(i) With reference to Fig. 1.1, explain why the gene of interest cannot be taken directly from chromosomal DNA.

....

....

....[2]

1. **Eukaryotic DNA** contains **introns** and
2. **bacterial** / **prokaryotic** host cells **do not have post-transcriptional modification / splicing to remove introns.**
3. **Non-functional protein may be synthesized** if introns are not removed.

(Max 2)

[L1]

(ii) Outline the additional processing required to yield the gene of interest prior to insertion into the plasmid.

....

....

....

....[3]

1. **Isolate the processed / mature mRNA** coding for human Growth Hormone
2. Use **reverse transcriptase** to **synthesize single-stranded cDNA** using the processed mRNA as template **(Reject: conversion)**
3. Use **DNA polymerase** to replicate the **single-stranded cDNA into double-stranded cDNA**.
4. **Addition of linker DNA** to the **ends** of the double stranded cDNA / Gene of interest.

(Max 3)

[L2]

(d) State one possible pair of gene markers present on the cloning site of the plasmid for the identification of desired clone in Fig. 1.1.

....

....

....[1]

1. **Ampicillin resistance gene** & **Tetracycline resistance gene**

OR

2. **LacZ gene / β-galactosidase gene** & **Ampicillin resistance gene**

(Max 1)

[L2]

(e) Outline the process for Stage E in Fig 1.1 using one of the gene markers in (d).

....

....

....

....[3]

1. **Replica-plate** the **master plate containing the bacterial clones** on two separate **agar plates containing ampicillin and tetracycline**.
2. **Bacterial clones / colonies** that grow on both antibiotic plates are **resistant to both antibiotics are non-recombinant**.
3. **Select for bacterial clones sensitive to the antibiotic of the gene marker in (d)** but **resistant to the other antibiotic** from the **master plate**.
4. Correct reference to **insertional inactivation**.

(Max 3)

OR

1. **Culture the bacterial clones / colonies** on **X-gal** medium with ampicllin.
2. **Bacterial clones / colonies** that appear **blue** are **non-recombinant** as LacZ gene is intact.
3. **Select for white bacterial clones** that are **recombinant**.
4. **LacZ gene is disrupted** due to **insertional inactivation**.

(Max 3)

[L2]

Fig 1.2 shows details of how stage A and B are carried out.

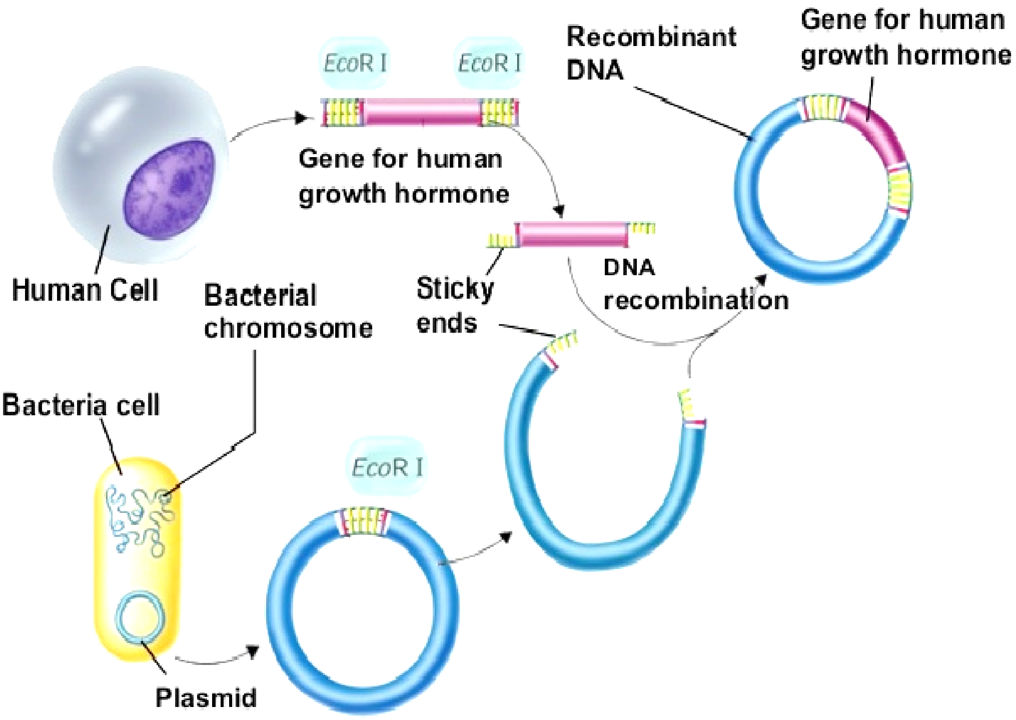


Fig 1.2

(f) A scientist commented that two different restriction enzymes should be used to isolate the gene for human growth hormone instead of using only *EcoRI* restriction enzyme. Explain the rationale behind his comment.

…………………………………………………………………………………………………

…………………………………………………………………………………………………

…………………………………………………………………………………………………[2]

1. The **gene** for human growth hormone **may insert into the plasmid in more than one orientation** / **OWTTE**
2. This may lead to **non-functional protein being synthesized** if the gene is inserted in the wrong orientation
3. Having two different restriction enzymes produces **two different sticky ends** to isolate the gene at the ends will ensure **uni-directional insertion** of the gene into the plasmid.

(Max 2)

[L3]

[Total: 13]

2 Restriction Fragment Length Polymorphism (RFLP) is an important application for comparative genetic analysis of normal and diseased individuals.

(a) Outline the key techniques used for RFLP analysis.

....

....

....

....

....[4]

1. **<u>Same</u>** set of **<u>restriction enzyme(s) to digest DNA</u>** samples.
2. **<u>Gel electrophoresis</u>** to **<u>separate DNA fragments</u>** of different **<u>length</u>**.
3. **<u>Southern Blotting</u>**
4. to **<u>transfer DNA fragments to nitrocellulose membrane</u>**.
5. **<u>Nucleic acid hybridization</u>** with **<u>labelled probes complementary to target sequences</u>**.
6. **<u>UV transillumination / Autoradiography</u>** (relevant to point 4 on type of probes) to **<u>visualized DNA fragment</u>** containing target sequences.

(Max 4)

[L1]

Fig. 2.1 shows the pedigree and southern blots of an RFLP locus that is closely linked with a disease phenotype. Affected individuals are shaded in the pedigree.

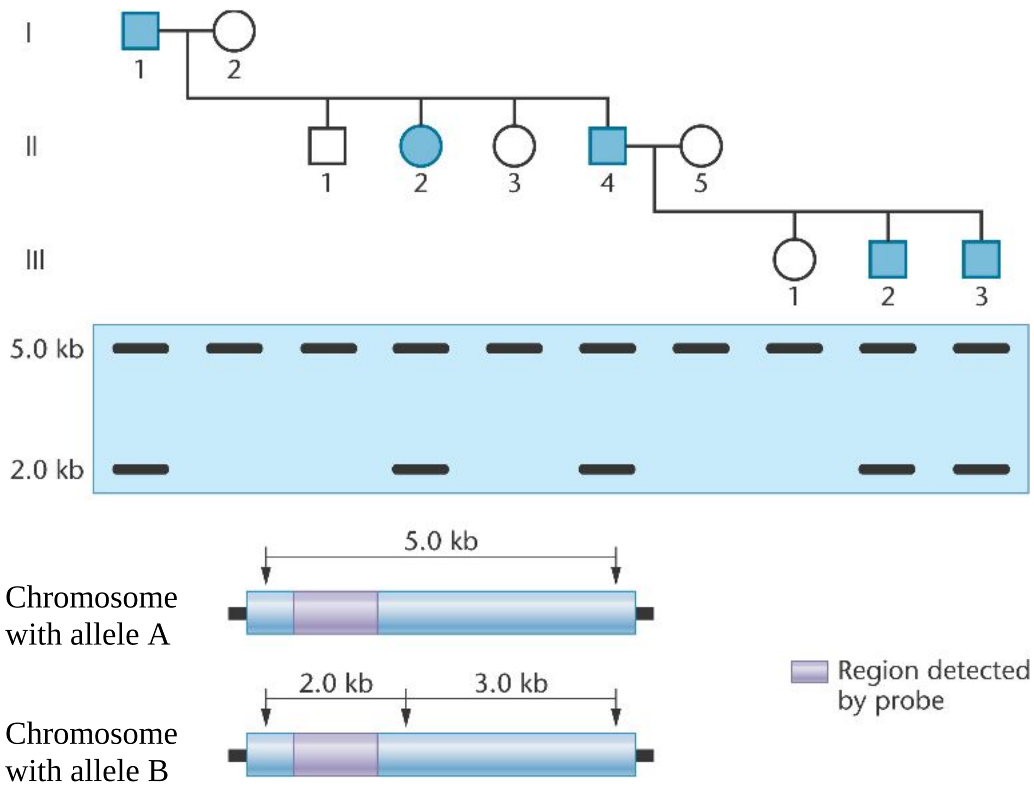


Fig. 2.1

(b) Explain the genetic basis of RFLP in comparative analysis in disease study.

……

……

…… [2]

1. **Normal and disease individual** have **difference(s) in genetic sequence**.
2. Upon **digestion with same set of restriction enzyme(s), different number and length of DNA fragment** will be produced.
3. Gel electrophoresis and nucleic acid hybridization with labelled probes will result in **different RFLP profile / band patterns** for normal and diseased individual.

[L2]

(c) With reference to Fig. 2.1,

(i) Identify the allele responsible for the disease.

……[1]

1. **Allele B**

[L2]

(ii) State the considerations for target region of the probe.

……

……

……[2]

1. **Same target sequence for allele A and B**.
2. **Disease allele** is **closely linked/associated** with **polymorphic RFLP locus**.
3. **Probe binds to different fragment lengths** for **normal and disease allele** / **OWTTE**.

[L3]

(iii) Explain if the disease allele is dominant or recessive.

……

……

……[2]

1. **Dominant**
2. **2kb fragment** associated with allele B is always present in **affected individuals** (I-1, II-2, II-4, III-2 and III-3).
3. In the **absence of allele B**, **individuals are normal**.

[L2]

(d) Briefly describe two other applications of RFLP.

………

………

……[2]

1. RFLP as **genetic markers in genomic/linkage/physical mapping**.
2. **DNA fingerprinting** for **paternity testing / forensic investigation / verification of poaching**.

[L1]

[Total: 13]

3 Fig. 3.1 shows how gene therapy using modified viruses can be carried out.

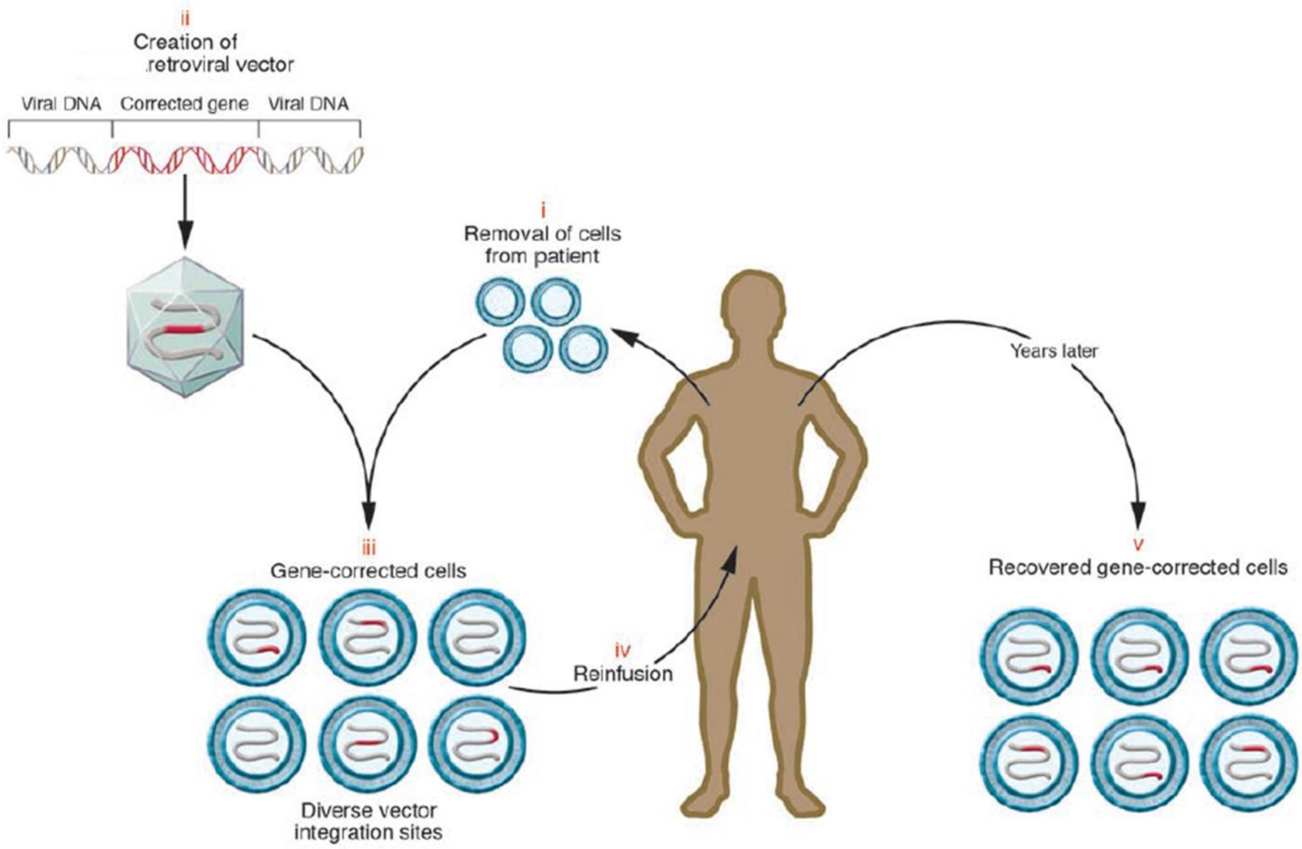


Figure obtained from Retroviral integration and human gene therapy The Journal of Clinical Investigation, Volume 117, Number 8

Fig. 3.1

Severe combined immunodeficiency (SCID) and X-linked SCID are two genetic diseases that can be treated by the gene therapy method shown in Fig. 3.1.

(a) State the genes that can be corrected for SCID and X-linked SCID.

SCID: ……………………………………………………………………………..

X-linked SCID: ………………………………………………………………………

[2]

S: State (CW), gene, corrected, X-linked SCID

C: Factual recall

ORE:

1. SCID: **Adenosine deaminase** gene Reject: ADA gene
2. X-linked SCID: Gene coding for a subunit **gamma c** (γc) of an **interleukin receptor**.

[L1]

(b) Explain how mutation in one of the genes cited in **(a)** would give rise to immunodeficiency.

……………………………………………………………………………………………………

……………………………………………………………………………………………………

…………………………………………………………………………………………………[2]

S: Explain why(CWs), mutations genes, (a), immuodeficiency

C:
- Cause and effect
- Adenosine deaminase gene mutation → Purines cannot be broken down → Toxic to cells
- Mutation in gene coding for a subunit gamma c (γc) of an interleukin receptor → HSC cannot convert to T-cells

ORE:
1. Adenosine deaminase (ADA) **breaks down purine**
2. hence without ADA, purines which **accumulate** is **toxic** to **T** and **B cells**, causing them to **die**.

   OR
3. γc subunit of IL receptors is needed to convert **hematopoietic stem cells to progenitors of T cells**,
4. hence without the the progenitors there will be **no derivation / generation** of **T-cells.**

(Points 1 & 2 have to be together & Points 3 & 4 have to be together.)

[L2]

Excerpt from the *The Journal of Clinical Investigation* review titled *Retroviral integration and human gene therapy*:

> "However, with these successes came the first serious adverse events in retrovirus based gene therapy. Three of the SCID-X1 patients treated by the French team developed a leukemia-like lymphoproliferative disease."

(c) Suggest why these patients developed leukemia-like lymphoproliferative diseases.

……………………………………………………………………………………………………

……………………………………………………………………………………………………

……………………………………………………………………………………………………[2]

S: Suggest why(CWs), patients, developed, leukemia-like lymphoproliferative diseases, retrovirus

C:
- Retroviruses integrate in the genome
- Protooncogene converted to oncogene
- Not tumour suppressor gene as two alleles need to be inactivated.

ORE:

1. **DNA** of modified **retroviruses integrate** into **host** cell **genome** / **insertional mutagenesis**.
2. Integration may cause **gain of function mutation** resulting in **protooncogene** converted to **oncogene**.
3. Integration may cause **loss of function mutation** in **2 alleles** of the **tumour suppressor gene**.

[L3]

In 1999, 18-year old Jesse Gelsinger died after suffering from a massive immune response following a clinical trial that administered adenoviral-based gene therapy.

(d) Besides adenovirus' ability to elicit a strong immune response, explain why adenovirus is also not favoured in the treatment of the genetic diseases cited in **(a)**.

……

……

………………………………………………………………………………………………………[2]

S: Explain why (CWs), besides, adenovirus, immune response, not favoured, treatment..genetic diseases..(a)

C:
- Cannot mention adenovirus' high immunogenicity
- Adenovirus does not integrate into host genome
- Limited duration of in vivo gene expression

ORE:
1. **Adenovirus does not integrate** into **host** cell **genome** after infection.
2. There is therefore **limited duration** of ***in vivo*** **gene expression / OWTTE**.

[L2]

(e) Explain why scientists prefer to isolate hematopoietic stem cells rather than T-cells from the patient for the gene therapy treatment shown in Fig. 3.1.

……

……

……

………………………………………………………………………………………………………[3]

S: Explain why (CWs), scientists, prefer, isolate, HSCs vs T-cells, gene therapy, Fig. 3.1

C:
- HSCs

ORE:
1. Gene targeting T-cells does **no**t result in **long-term expression** of the corrected **genes**, since most of these cells **die** rather than self-renew.
2. Gene targeting haematopoietic stem cells result long-term expression of the corrected genes as these cells have the capability to **self-renew**
3. and **differentiate** to give rise to more **T-cells** /
4. and are **multipotent stem cells**.

[L2]

Viruses are not the only vectors that are utilised to deliver genes into patients' host cells.

(f) State one other type of vector that is not viral.

……

………………………………………………………………………………………………………[1]

S: State (CW), one, other type, vector, not viral

C:
- Factual recall

ORE:

1. **Liposomes**
2. **Gene gun**

(Max 1)

[L1]

(g) Discuss two ethical implications of gene therapy.

……

……

……

……[2]

S: Discuss (CW), two, ethical implications, gene therapy

C:
- Discursive question

ORE:

1. Philosophical perspective on **morality** of changing genetics / OWTTE.
2. **Germ-line gene therapy** will alter offspring genetics, we may not have the **right to alter** our **children's genes** / OWTTE.
3. Those involved in the research related to **germ-line gene therapy** regularly **create and destroy embryos** as a part of their research. There are objections to killing embryos used for research in gene therapy as **human life** may be considered to have begun at conception.
4. **Genetic determinism / Definition of 'normal' and 'disability'**. The decision of what is normal and what is a disability should not rest on just a small group of individuals or medical professionals. The question of whether disabilities are considered disease and whether these need to be cured or prevented requires thorough consideration.
5. **Patenting** to **develop drug/process at the expense of human life** / OWTTE.
   Reject all social/religious points.
   (Max 2)

[L2]

[Total: 14]

4 Planning question

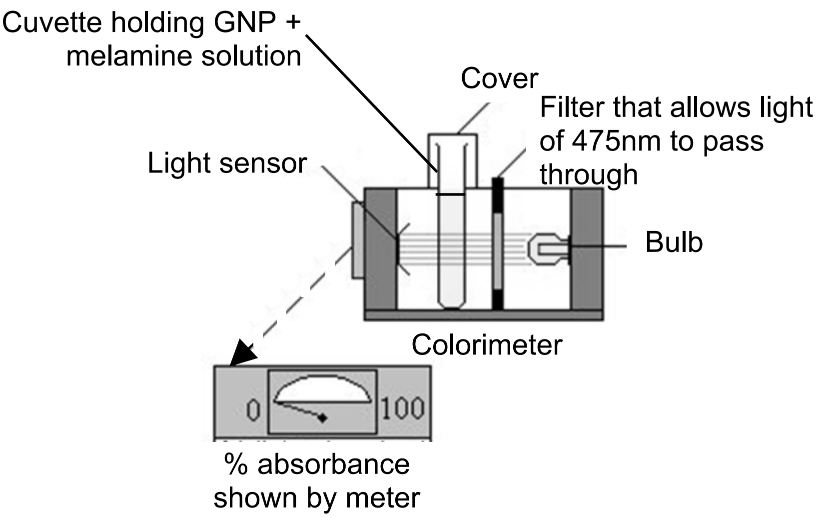
In 2008, contaminated milk and infant formula with melamine in China resulted in over a thousand babies hospitalised and caused several deaths. Melamine with its large number of nitrogen atoms, was unethically added to foods to mimic proteins detected through conventional measurements for nitrogen.

A research group found a way to detect the presence of melamine via a colour indicator. They attached cyanuric acid to gold nanoparticles to produce functionalized gold nanoparticles (GNP) that are red, the cyanuric acid component of GNP detects the melamine through hydrogen-bonding. Upon binding to melamine, the GNP changes from red to blue, giving a clear signal that there is contamination.

GNP (red) + Melamine → GNP-Melamine complex (blue)

The intensity of the blue colour after a set time interval of 5 minutes is a measure of the concentration of the melamine present in a specimen.

The test involves addition of 1 cm^3 of solution (to be tested) to 0.1 cm^3 of GNP in a cuvette (glass container), followed by colourimetric measurement of absorbance (% absorbance) at a specified wavelength. It is known that the blue wavelength is 475 nm.



You are required to plan, but not carry out, an investigation to determine the lowest concentration of melamine detectable by GNP indicator.

Your planning must be based on the assumption that you have been provided with the following equipment and materials which you **must** use:

- 1 % GNP
- Colourimeter
- 6 x Cuvettes (container for colourimeter measurement)
- 6 x Test-tubes
- Test-tube rack
- 2 x 10 cm^3 syringes
- 2 x 1 cm^3 syringes
- Micropipette
- 15 cm^3 of 1% melamine solution labelled M
- 50 cm^3 of buffer solution
- Marker pen
- Stopwatch
- Safety googles
- Disposable gloves

Your plan should have a clear and helpful structure to include:

- an explanation of theory to support your practical procedure
- a description of the method used including the scientific reasoning behind the method
- proposed layout of results tables with clear headings and labels
- the correct use of technical and scientific terms

[Total: 12]

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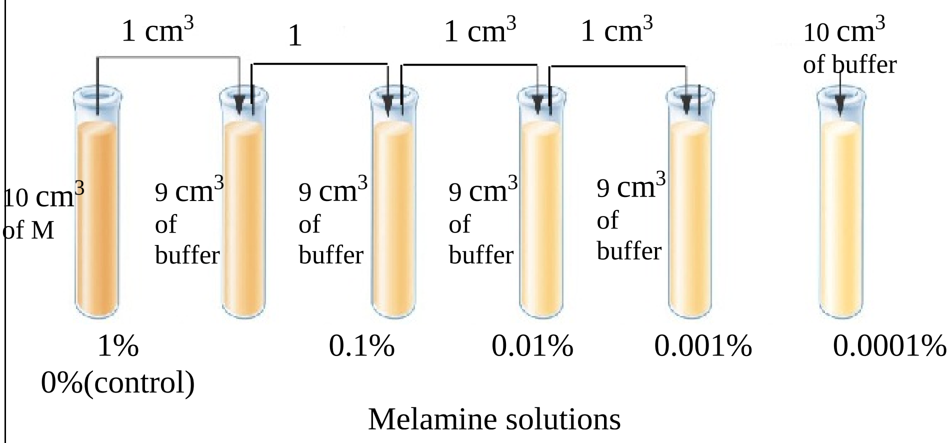
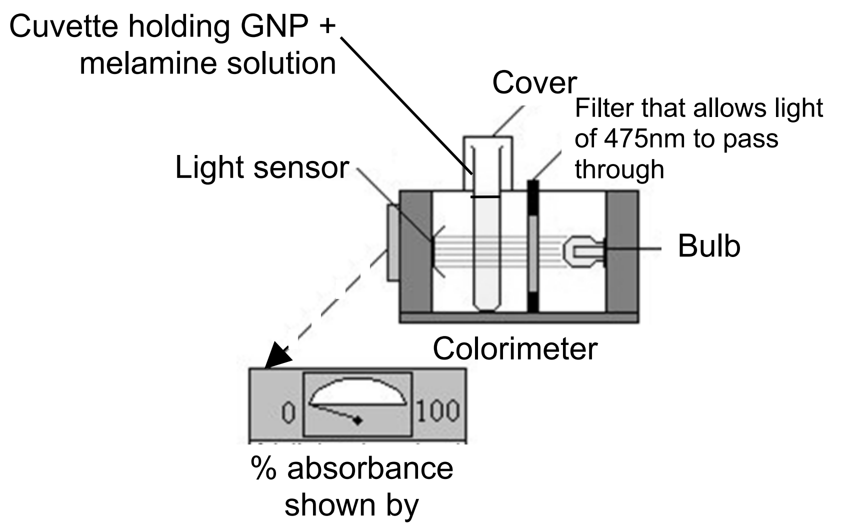
Section	Marking Points	
Aim, Hypothesis, Introduction [Max 2]	1. **Melamine may mimic protein in infant formula**.	**1**
	2. **Presence of melamine can be detected by a colour change of GNP from red to blue**	**1**
	3. **Melamine binds to cyanuric acid component of GNP through hydrogen-bonding**, is the basis for detection.	**1**
	4. The presence of melamine contaminant could be **measured by a colorimeter** that measure **% absorbance at specifc wavelength**	**1**

<table>
<tr><td></td><td>(475nm),</td><td></td></tr>
<tr><td rowspan="2">Variables [1]

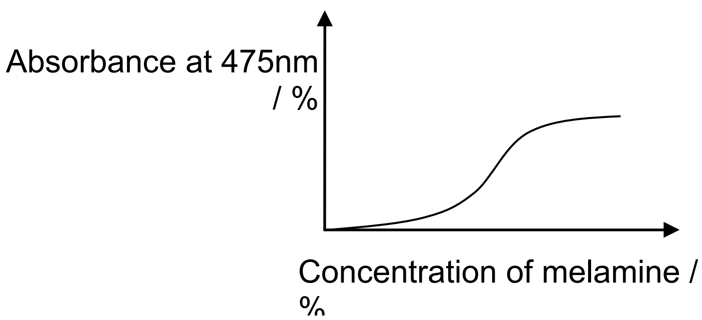
Both Independent & Dependent variable must be correct.</td><td>5. Independent variable: <u>Concentration of melamine / % (1%, 0.1%, 0.01%, 0.001%, 0.0001%)</u></td><td rowspan="2">1</td></tr>
<tr><td>6. Dependent variable: <u>% absorbance of solution at specific wavelength (475nm)</u> after <u>5 mins</u></td></tr>
<tr><td rowspan="4">Apparatus & Procedures [Max 4]

Note: rationale must be included for award of marks.</td><td>7. <u>Carry out serial dilution of 1% melamine solution using the buffer</u> to reduce the concentration of melamine solution by a factor of 10 between each of the four successive dilutions, to give <u>0.1%, 0.01%, 0.001% and 0.0001% (Description / Table of dilution accepted)</u></td><td>1</td></tr>
<tr><td>8. Clear labelled diagram of experimental setup





</td><td>1</td></tr>
<tr><td>9. <u>Negative Control contains buffer only</u>. All other conditions remain the same. This shows that colour change of GNP indicator to blue only occurs in the presence of melamine.</td><td>1</td></tr>
<tr><td>10. Data points [Five melamine concentration]: Table of preparation must be shown
<table>
<tr><td>Concentration of melamine/% (no need to consider s.f.)</td><td>1.0000</td><td>0.1000</td><td>0.0100</td><td>0.0010</td><td>0.0001</td><td>0.0000</td></tr>
<tr><td>Melamine solution to be diluted</td><td></td><td>1.0000</td><td>0.1000</td><td>0.0100</td><td>0.0010</td><td></td></tr>
<tr><td>Volume of melamine solution to be diluted / cm³</td><td>10.0</td><td>1.0</td><td>1.0</td><td>1.0</td><td>1.0</td><td>0.0</td></tr>
<tr><td>Volume of buffer to be added / cm³</td><td>0.0</td><td>9.0</td><td>9.0</td><td>9.0</td><td>9.0</td><td>10.0</td></tr>
</table>
</td><td>1</td></tr>
</table>

<table>
<tr><td></td><td>11. Controlled variable: Volume of melamine solution tested kept constant to 1cm^3</td><td>1</td></tr>
<tr><td></td><td>12. Controlled variable: Volume of GNP indicator used kept constant to 0.1 cm^3</td><td>1</td></tr>
<tr><td></td><td>13. Controlled variable: Solution reaction time to GNP indicator kept constant to 5 mins.</td><td>1</td></tr>
<tr><td></td><td>14. After 1 cm^3 of melamine solution is added to 0,1 cm^3 of GNP indicator and stand for 5 mins in the cuvette.</td><td>1</td></tr>
<tr><td></td><td>15. [Colorimeter Calibration] Fill another cuvette with buffer/control to calibrate the colorimeter zero absorbance reading.</td><td>1</td></tr>
<tr><td></td><td>16. Measure the % absorbance of solution using colorimeter.</td><td>1</td></tr>
<tr><td></td><td>17. Repeat Step 11-16 for other concentrations of melamine solutions.</td><td>1</td></tr>
<tr><td></td><td>18. [Data collection]: Record the % absorbance of the solutions in a table.</td><td>1</td></tr>
<tr><td></td><td>19. [Reliability]: Perform 2 repeats to obtain 3 replicates → ensure reliability [See point 23-24]</td><td>1</td></tr>
<tr><td>Safety & Precautions [1]</td><td>20. Specify at least <u>ONE</u> valid risk and corresponding precaution:
• [Risk] Use of glassware during preparation of melamine serial dilution, exercise caution + [Precaution] prevent breakage of glassware to avoid cut injuries.
• [Risk] Wet hands could risk electrocution when switching on colorimeter + [Precaution] Dry hands before switching on colorimeter.
• [Risk] Melamine is poisonous + [Precaution] wear gloves and protective googles to prevent exposure.
• AVP</td><td>1</td></tr>
<tr><td>Results [Max 2]</td><td>19. [Table of results] must include:
• Independent variable with unit
• Dependent variable with unit
• Processed data with unit
• Correct trend (in agreement with hypothesis – lower absorbance at lower concentration of melamine)
[T] <u>Table showing intensity of colour of extract</u>
<table>
<tr><td rowspan="2">Concentration of melamine solution / %</td><td colspan="4">Absorbance at 475nm / %</td></tr>
<tr><td>Replicate 1</td><td>Replicate 2</td><td>Replicate 3</td><td>Average</td></tr>
<tr><td>1.0000</td><td></td><td></td><td></td><td></td></tr>
<tr><td>0.1000</td><td></td><td></td><td></td><td></td></tr>
<tr><td>0.0100</td><td></td><td></td><td></td><td></td></tr>
<tr><td>0.0010</td><td></td><td></td><td></td><td></td></tr>
<tr><td>0.0001</td><td></td><td></td><td></td><td></td></tr>
<tr><td>0.00 (control)</td><td></td><td></td><td></td><td></td></tr>
</table></td><td>1</td></tr>
<tr><td></td><td>20. [Graph] must include:
• Independent variable on x-axis labelled with unit
• Dependent variable on y-axis labelled with unit
• Line graph (Increased % absorbance with increasing melamine concentration)

</td><td>1</td></tr>
<tr><td>Interpretation [1]</td><td>21. [Correlate with graph and hypothesis]:
• As shown by the graph, as the concentration of melamine increases, average % absorbance at 475nm increases, given a</td><td>1</td></tr>
</table>

	fixed amount of time of exposure / Vice versa	
	22. **Lowest concentration of melamine that could be detected can be determined based on the graph / comparison against control (same result).**	
	23. Further serial dilutions may be required to determine lowest concentration of melamine that could be detected.	
Correct scientific terms + Reasoning [1]	24. Use **appropriate terms** within answer (e.g. **hydrogen bonding, sensitivity, absorbance** etc.)	**1**

- **Note: Reference to SPA Planning Qn on Membrane structure (SPA Planning Booklet pg)**

Section B: Free-response question

Answer **all** questions. Answer each part on a **separate** piece of paper.

Write your answers on separate answer paper provided.
Your answer should be illustrated by large, clearly labelled diagrams, where appropriate.
Your answer must be in continuous prose, where appropriate.
Your answer must be set out in sections **(a)**, **(b)** etc., as indicated in the question.

5 **(a)** Compare between Genomic and cDNA libraries used in protein production. [6]

Basis of Comparison	Genomic Library	cDNA libraries
Similarities [Max 3]	1. Both store genetic information	
	2. Both requires the use of vectors	
	3. Both require ligation of genetic sequence to vector using DNA ligase	
	4. Both requires storage in host cells.	
Difference [Max 3]		
5. Scope of genetic information	Entire genome inclusive of introns and exons	Expressed part of the genome
6. Starting material for library construction	Genomic DNA	Processed mRNA
7. Cloning Vector	Bacteriophage, Cosmid, BAC, YAC,	Plasmid and λ phage
8. Enzymes involved	Restriction enzymes, DNA ligase, DNA polymerase	Restriction enzymes, DNA ligase, DNA polymerase and reverse transcriptase
9. Ease of library screening	Difficult to locate gene of interest as single gene may be dispersed over several clones.	Easier as gene are isolated whole.
10. Expression of eukaryotic gene in prokaryotic system	Unlikely to produce functional protein due to the presence of introns.	Functional protein can be produced

11. Nature of genetic information stored	Same throughout the life of the cell / regardless on the type of cell	May differ at different time in the life of the cell / dependent on the type of cell
12. Presence of introns	Present	Absent
13. Presence of regulatory sequences	Present	Absent

[L3]

(b) Explain the advantages and limitations of PCR. [6]

Advantages (Max 3]:

1. PCR is **highly specific**, only sequenced flanked by **forward and reverse primers** are replicated.
2. PCR can be performed **in vitro** without the use of cells.
3. PCR is a **cheaper technique as compared to cloning** because no need to culture and maintain large quantities of host cells.
4. PCR is **faster than cloning** in replication of DNA.
5. Only **minute amounts of DNA** is required as **starting material/ for amplification**.

Limitations (Max 3):

6. Process is **not error-free** due to **absence of proofreading activity** in **Taq polymerase**.
7. Possible **contamination from non-template DNA** if primers sequence are not specific.
8. DNA sequences **flanking target sequences** to be amplified **must be known** to enable **synthesis of primers**.
9. **PCR cannot substitute gene cloning** in cells for **longer DNA sequences**.

(c) Discuss the pros and cons of genetically-modified crop plants. [8]

SC	OR	E
CW: Discuss QKW: pros & cons, genetically-modified crop plants PRO	• Crop yield increase	1. Genetic engineering on plants can **enhance crop yields**.
PRO	• May bypass seasonal restrictions	2. Genetic engineering may permit crops to **grow outside** their **usual location** / **season** so that people have **more food**.

PRO	• Enhance nutritional content	3. Genetically modified crop plants can also be **enhanced** with a certain **nutritional content** (e.g. Golden rice) so that people are **better fed/OWTTE**.
PRO	• Pest-resistant crops	4. Genetically modified crop plants can be more **pest-resistant** (e.g BT corn),
PRO	• Lower cost	5. and this will **lower cost** as **pesticide** usage will be **reduced/avoided**.
PRO	• Less pollution	6. As pesticide usage is reduced/avoided, there will be **less damage/pollution** to the **environment**.
PRO	• Drought-resistance	7. Genetically modified crop plants can be more **drought-resistant**, and this will increase crop yield for farmers.
PRO	• Avoid costly irrigation	8. Drought-resistant crops can also help farmers **avoid** installing **costly irrigation** systems to ensure sufficient water is provided.
PRO	• Profits increase → consumer cost drops	9. As farmers' cost is reduced, their **profits increase** / **consumer cost** may also **reduce**.
PRO	• Increased shelf-life	10. **Shelf-life** of **crops** can be **increased** **Flavor Savr PG gene** example must be given
CON	• More invasive plants • Superweeds	11. The introduced gene(s) may be **transferred by pollen** to **wild relatives** whose hybrid offspring will become more **invasive** and hence become **'superweeds'**.
CON	• Cost involved in removing superweeds	12. This may lead to **additional cost for** the **removal** of such **'superweeds'.**
CON	• Plant diversity compromised	13. 'Superweeds' may also **reduce plant biodiversity** by **out-competing** natural plants
CON	• Organic farms may be compromised	14. The introduced gene(s) may be **transferred by pollen** to **unmodified plants** growing on a farm with **'organic'** certification, hence losing organic certification.
CON	• Toxic components	15. The modified plants will be a direct hazard to humans, domestic **animals** or other beneficial animals by being **toxic.** OR For instance, the herbicides that can now be used on the crop will itself **leave toxic residues** in the crop, which will be toxic to humans who consume them.

		OR **Disrupt ecological systems** e.g. Monarch Butterflies affected by BT.
	• Damage environment	16. The toxic residues/ overused pesticide may also affect and cause **unintended damage** to the **environment** by causing pollution to the surroundings.
	• Allergies	17. The modified plants will be a direct hazard to humans, domestic animals or other beneficial animals by being **producing allergies** upon consumption.
	• Farmers pay royalties	18. Farmers may have to **pay royalties** to sow the crops. OR 18. Some companies have extended their **patents on genetically engineered seeds** to prevent farmers from re-sowing the seed from these genetically engineered crops.
	• Higher consumer cost	19. The cost that farmers have to pay for the royalties may be passed on the **consumers** who may have to **pay higher prices** for them.
	• Vulnerability to diseases	20. **Genetically identical** and would be **equally vulnerable to disease.**

(Max 8, Maximum 4 pros and 4 cons)

[L2]

[Total: 20]

END OF PAPER